

Composition $\text{Mockingbird} \Rightarrow Q \circ B$ or $B' \circ A$ if $f, g, g(f(x))$ i.e. reversed composition
Mockingbird $\Rightarrow \omega \circ \lambda x.xx$ i.e. applying a function to itself
Omega $\Rightarrow Q \circ \omega$ i.e. a non-terminating combinator which has no head normal form (let alone β -normal form) i.e. its unsolvable
Warbler $\Rightarrow W \circ \lambda f.x.fxx$ i.e. argument duplicator; e.g. $W(\lambda x)x = x \circ x = 2x$
Cardinal $\Rightarrow C \circ \lambda fxy.fxyx$ i.e. flip
Viero $\Rightarrow V \circ \lambda xyx.zxy$ i.e. pair constructor
Starling $\Rightarrow S \circ \lambda f g x.f(x)(g(x))$ provides context of x to f, g ; think: reader monad
Psi $\Rightarrow \Psi \circ \lambda f g x.f(gx)(gy)$ processes two arguments before combining them;
WC (WC length) will add lengths of two lists
Recursion > Fixpoint Combinators
Fixed-Point of a function is where it returns the unchanged input i.e. for any x if $f(x) = x$ then x is a fixed-point e.g. double $0 = 0$
Fixed-Point Combinator fix is such that: if f has any fixed-points, then $\text{fix } f$ is one of those fixed-points, i.e. $f(\text{fix } f) = \text{fix } f$
These fixed point operators can be used to implement recursion in λ -calculus, e.g. factorial function $\text{fact} \Rightarrow \lambda n.f$ (if $\text{isZero } n$ then 1 else $n \times \text{fact}(n-1)$)
Now we can simply extract $\text{fact} \Rightarrow \lambda n.f$ (if $\text{isZero } n$ then 1 else $n \times (f(n-1))$) fact
The inner part we can call F i.e. $F \Rightarrow \lambda n.f$ (if $\text{isZero } n$ then 1 else $n \times (f(n-1))$)
So now its $\text{fact} = F \text{ fact}$
Now we use fix to re-write it as $\text{fact} = \text{fix } F$
Since $\text{fix } F = F(\text{fix } F)$ and $\text{fix } F = F(\text{fix } F)$ we get $\text{fact} = F \text{ fact}$ i.e. our original definition

Y-Combinator $\Rightarrow Y \circ \lambda B.\omega(C \circ B)\omega(\lambda f.(xx)(\lambda x.fxx))$
 $Y f \Rightarrow f(Y f)$ i.e. both β -reduce to the same common term
However $f(Y f)$ may not in general β -reduce to $Y f$
Z-Combinator $\Rightarrow Z \circ \lambda f.(xx.f)(y.xxy)(\lambda x.f(y.xxy))$
 $Z f \times f(f(Z f)x)$ i.e. next argument is defined explicitly
Y-combinator of **Y-Combinator**, which will work for strict languages (Y-Combinator will stack overflow)
Turing (U) and Theta (Θ) combinators:
 $U \circ \lambda xxy.y(xyxy) \circ \Theta \circ \lambda x$
These mirror the definitions of Mockingbird and Omega combinators
 $\Theta f \Rightarrow f(\Theta f)$
i.e. (unlike Y-combinator) it actually β -reduces

Encoding datatypes in λ -calculus
Church Booleans $\Rightarrow$ encoding $B = \{\text{true}, \text{false}\}$
 $\text{true} \equiv \lambda x.\lambda y.x$
 $\text{false} \equiv \lambda x.\lambda y.y$
Basic Logical Operators:
if p then x else y $\Rightarrow$ $\text{if } p \text{ then } x \text{ else } y \Rightarrow \text{if } p \text{ then } x \text{ else } y$
 $\text{not} \equiv \lambda xxy.xxy$
or $p \vee q \Rightarrow \text{if } p \text{ then } p \text{ else } q$
and $p \wedge q \Rightarrow \text{if } p \text{ then } q \text{ else } p$
Common Predicates:
 $\text{isZero} \equiv \lambda n.n(\lambda x.\text{false}) \text{true} \Rightarrow$
 $\text{ifZero} \equiv \lambda nxy.n(\lambda x.y)x$
 $(s) \equiv \lambda mn.\text{isZero}(m-n)$
Church Numerals $\Rightarrow$ encoding $\mathbb{N}$
Church numerals: $\bar{n} \equiv \lambda f.\lambda x.f^n(x)$
 $\bar{n} \equiv \lambda f.\lambda x.f^n(x)$ is a function that returns the n th composition of f
Basic Arithmetic Operators:
Successor $\text{succ} \equiv \lambda nfx.f(nfx)$
Addition $\text{plus} \equiv \lambda xy.x + y$
 $\text{plus} \equiv \lambda xy.x + y$
Multiplication $\text{mult} \equiv \lambda xy.x \times y$
 $\text{mult} \equiv \lambda xy.x \times y$
Exponentiation $\text{pow} \equiv \lambda be.b \rightarrow^n e$
 $\text{pow} \equiv \lambda be.b \rightarrow^n e$
Predecessor $\text{pred} \equiv \lambda n$
 $\text{pred} \equiv \lambda n$

Subtraction $\text{sub} \equiv \lambda mn.m \text{ pred } n$
 $\text{sub} \equiv \lambda mn.m \text{ pred } n$
State is a partial function from variables to numbers
 $(\text{State} = V \mapsto \mathbb{N})$ such that $s(x)$ is defined for finitely many x .
E.g.
 $s_1 = \{x \mapsto 4, y \mapsto 5, z \mapsto 6\}$
Our small-step semantics for While will be defined using configurations of the form (E, s) , (B, s) and (C, s) . This expresses that we evaluate E , B and C with respect to state s , and may modify s in the process.
State Update
We define the state update operation $[v \mapsto n]$ on states as follows
 $s[v \mapsto n](u) = n$ if $u = v$
 $= s(u)$ otherwise
where v and u are variables, and n is a number.
For example, with $s_1 = \{x \mapsto 4, y \mapsto 5, z \mapsto 6\}$
we have that $s_1[y \mapsto 88] = \{x \mapsto 4, y \mapsto 88, z \mapsto 6\}$
Expressions
 $\rightarrow_e \subseteq (\text{Exp} \times \text{State}) \times (\text{Exp} \times \text{State})$
Inductive Principles
Induction Over Terms
Induction for Nat
The set Nat can be defined by the following grammar:
 $N \in \text{Nat} ::= \text{zero} \mid \text{succ}(N)$
These give rise to the following induction principle, for any $P \subseteq \text{Nat}$:
 $P(\text{zero})$
 $\wedge$
 $\forall n \in \text{Nat}. [P(n) \Rightarrow P(\text{succ}(n))]$
 $\implies$
 $\forall n \in \text{Nat}. [P(n)]$
Induction on the Syntax for Binary Trees
Binary trees defined by the following grammar:
 $\text{bTree} \in \text{Tree} ::= \text{leaf} \mid \text{Node}(\text{bTree}, \text{bTree})$
Inductive principle for binary trees:
 $P(\text{leaf})$
 $\wedge$
 $\forall \text{bt1}, \text{bt2} \in \text{Tree}. [P(\text{bt1}) \wedge P(\text{bt2}) \Rightarrow P(\text{Node}(\text{bt1}, \text{bt2}))]$
 $\implies$
 $\forall \text{bt} \in \text{Tree}. [P(\text{bt})]$
Principle of Induction for Simple Expressions
 $E \in \text{SimpleExpr} ::= n \mid E +_s E' \mid E \times_s E'$
Inductive Principle
 $\forall n \in \mathbb{N}. P(n)$
 $\wedge$
 $\forall E1, E2 \in \text{SimpleExpr}. [P(E1) \wedge P(E2) \Rightarrow P(E1 +_s E2)]$
 $\wedge$
 $\forall E1, E2 \in \text{SimpleExpr}. [P(E1) \wedge P(E2) \Rightarrow P(E1 \times_s E2)]$
 $\implies$
 $\forall E \in \text{SimpleExpr}. [P(E)]$
Induction Over Relations
 $\text{Data via } \rightarrow$ resulting in a big/small-steps semantics
 $\text{data} \in (\mathbb{N} \times \mathbb{N} \times \mathbb{N}) \times \mathbb{N}$
 $\text{M1} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M2} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M3} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M4} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M5} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M6} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M7} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M8} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M9} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M10} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M11} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M12} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M13} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M14} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M15} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M16} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M17} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M18} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M19} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M20} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M21} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M22} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M23} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M24} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M25} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M26} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M27} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M28} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M29} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M30} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M31} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M32} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M33} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M34} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M35} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M36} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M37} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M38} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M39} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M40} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M41} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M42} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M43} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M44} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M45} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M46} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M47} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M48} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M49} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M50} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M51} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M52} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M53} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M54} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M55} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M56} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M57} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M58} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M59} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M60} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M61} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M62} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M63} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M64} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M65} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M66} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M67} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M68} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M69} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M70} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M71} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M72} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M73} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M74} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M75} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M76} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M77} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M78} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M79} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M80} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M81} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M82} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M83} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M84} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M85} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M86} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M87} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M88} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M89} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M90} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M91} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M92} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M93} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M94} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M95} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M96} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M97} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M98} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M99} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M100} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M101} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M102} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M103} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M104} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M105} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M106} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M107} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M108} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M109} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M110} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M111} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M112} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M113} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M114} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M115} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M116} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M117} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M118} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M119} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M120} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M121} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M122} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M123} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M124} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M125} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M126} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M127} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M128} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M129} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M130} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M131} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M132} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M133} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M134} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M135} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M136} \equiv \{(i, \text{conf}, \text{data}) \mid \text{data} \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \times \mathbb{N}\}$
 $\text{M137} \equiv \{($